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Field confirmation of genetic variation in soybean transpiration response to vapor pressure deficit and photosynthetic compensation

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ABSTRACT

Plants with limited transpiration rate (TR) under high vapor pressure deficit (VPD) offer the potential to conserve soil water and thus decrease the occurrence of soil water deficit. Genetic variability in TR response to VPD has been observed in the greenhouse for soybean (*Glycine max* (L.) Merr.) genotypes related to PI416937, but these differences have yet to be measured in the field. The objective of this study was to observe under field conditions leaf gas exchange properties of PI416937 in comparison to nine other genotypes to determine if it expressed limited TR at high VPD. Genotypic differences in stomatal conductance measurements (a proxy for TR) matched those obtained under controlled environment conditions. Genotypes varied from no stomatal response to VPD, to strong negative responses resulting in full stomata closure at ~4 kPa. There was a greater proportional genetic variability in stomatal conductance in the field (75% at high VPD) than was observed in the greenhouse, but this variation was correlated with greenhouse TR. However, photosynthesis was considerably limited in genotypes that had a stomatal response to VPD. Although field differences in photosynthetic capacity among genotypes were not correlated with greenhouse measurements, there was sufficient genetic variation to allow the possibility of selection of high photosynthetic capacity to overcome about a 34% decrease in stomatal conductance. Thus, a targeted breeding program to combine the water conserving TR-VPD response with increased photosynthetic capacity has the potential to increase soybean yields in field water-deficit environments.

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1. Introduction

Several soybean [*Glycine max* (L.) Merr.] genotypes, including PI416937, have been identified as having a maximum or constant transpiration rate (TR) at high vapor pressure deficit (VPD) (Fletcher et al., 2007; Sadok and Sinclair, 2009a,b). When this trait was included in a simulation of soybean production across the United States, it resulted in water conservation and marked yield increases when short, dry periods occurred in the growing season (Sinclair et al., 2010). The hypothesized mechanism for the maximum-transpiration-rate trait is a leaf level hydraulic limitation that results in limited water transport and decreased stomatal conductance (g_{H2O}) at high VPD (Sinclair et al., 2008).

Decreased stomatal conductance (g_{H2O}), however, often negatively impacts photosynthetic rats (A), and thus can depress potential crop yield. Greenhouse studies of a range of soybean genotypes, however, demonstrated that while the constant TR at high, increasing VPD trait was associated with decreased g_{H2O} and lower photosynthetic rates, genotypes with higher photosynthetic capacity could compensate for the stomatal limitation effect (Gilbert et al., 2011). Based on these observations, Gilbert et al. (2011) proposed a breeding scheme to separate genetic variation in transpiration from photosynthetic capacity by measuring the components of intrinsic water use efficiency.

Intrinsic water use efficiency (WUE_{intr}) is the ratio between A and g_{H2O}, and is widely used as an index for beneficial characteristics with respect to plant water use. As the relationship between these two variables is curvilinear for soybeans, variation in g_{H2O} and/or photosynthetic capacity independently affect the value of WUE_{intr}. Thus, to use information on WUE_{intr} for breeding water-conserving soybeans, it is necessary to decouple which factor is causing variation in WUE_{intr}. Increases in photosynthetic capacity – photosynthetic differences between genotypes at a constant g_{H2O} – is particularly desirable, as it allows a genotype with decreased g_{H2O} to have similar photosynthetic rates to genotypes with higher g_{H2O}, and thus the potential for similar productivity. The contribution of variation in photosynthetic capacity to changes in WUE_{intr} was

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defined as WUE_{PC} (Gilbert et al., 2011). For a range of greenhouse-grown soybean genotypes, high WUE_{PC} was not correlated with reduced g_{H2O}, and sufficient variation in WUE_{PC} existed among the genotypes to compensate for a 20% reduction in g_{H2O}. Furthermore, these patterns were maintained under drought and variation in temperature.

All of the studies cited above have been conducted on potted plants in greenhouses. It remains to be demonstrated that: (1) genotypic variation in the response of TR to high VPD relationship is expressed in the field, (2) this variation is related to changes in g_{H2O}, and (3) that photosynthetic capacity differences are present among genotypes. The objective of this study was to observe these gas exchange properties on plants grown under field conditions, and assess if controlled environment observations apply to field-grown plants.

2. Material and methods

2.1. Field study

Soybeans genotypes were selected to represent the two broad physiological groups distinguished by Sadok and Sinclair (2009a). The first group was genotypes that had been observed in controlled environments to have constant linear increases in TR over the whole range of VPD: N01-11136, N96-7031 and PI471938. The second group of genotypes had initially linear increases in TR with increasing VPD, followed by a constant, or slightly increasing, TR at high VPD: A5959, Benning, Biloxi, PI416937, G00BP-59, G00BP-78 and G00BP-160. Genotypes in the latter group were selected to result in a broad range of maximum TR rates at high VPD (Table 1).

These genotypes were sown on 8 June 2010 in a field at the Central Crops Research Station near Clayton, North Carolina (35.669°N, 78.483°W). The soil was Bibb loamy sand (coarse-loamy, siliceous, active, acid, thermic Typic Fluvaquents). Fertilizer of 27 kg ha^{-1} of N, 12 kg ha^{-1} of P, and 134 kg ha^{-1} of K was applied to the soil prior to sowing. Each plot consisting of three rows of 3.05 m in length, spaced 0.97 m apart, was sown at a density of 26 seeds per meter and the plots were separated at the end of the row by 0.90 m pathways. Three randomly positioned replicate plots were sown for each genotype. The plots were irrigated when necessary to avoid soil water deficits.

2.2. Gas exchange measurements

Gas exchange measurements were made in the field on attached leaves once the plants had reached vegetative stage V3 to V4. On 12–15th July between 09:30 and 16:00 EST, gas exchange measurements were made on sun-exposed center leaflets of the second or third trifoliolate leaf. The g_{H2O} and A of a 6 cm^2 area of the leaflet was measured using an infra-red gas exchange system (Model 6400, LI-COR, Lincoln, NE). The chamber was positioned on a lateral half of the leaflets so that the midvein was not contained within the measurement area in the chamber. Light was provided in the chamber with red–blue LED light source to expose the leaflet to 2000 μmol m^{-2} s^{-1} PPFD. The air flow volume in the chamber was 600 μmol s^{-1} air flow rate and the atmosphere in the chamber was mixed by a fan.

The CO_2 mixer of the gas exchange system malfunctioned on the first day necessitating that chamber CO_2 fraction was a function of ambient CO_2 minus the leaf photosynthetic delta, and ranged from 328 to 393 μmol mol^{-1}. An upwind buffer volume was used to decrease variability in incoming CO_2. Repairs allowed full control of the chamber CO_2 fraction on subsequent days and it was set at 380 μmol mol^{-1}. As measurements on the first day provided valid g_{H2O} these were used in the analysis, however only photosynthetic data from the second to fourth days were used.

Leaf temperature and chamber humidity were allowed to adjust to ambient conditions; however the chamber-block temperature was not allowed to exceed 34 °C. Ambient conditions had sufficiently high irradiance for equilibration of leaf gas exchange in the chamber to occur within 2 min so that there was little further change in photosynthesis rate. A single leaf of each genotype, from each plot was measured in rotation, over the course of the day, resulting in between 14 and 23 measurements per genotype over the four days.

During measurement of each leaflet ambient temperature and VPD in the shade was measured using a digital psychrometer (Model SAM990DW, General Tools) accurate to 5% RH. Since the conditions in the gas exchange chamber could not practically match the prevailing external temperature, VPD, and canopy and boundary layer conditions, measurements of transpiration rate in the chamber were not a reliable estimate of water flux from ambient leaves (McDermitt, 1990). However, leaf stomatal conductance

Table 1
Equations for transpiration (TR) to vapor pressure deficit (VPD) responses measured in past greenhouse studies (Gilbert et al., 2011), and the origin of the ten soybean genotypes for which field VPD experiments were undertaken. Fletcher et al. (2007) provided VPD response data for PI471938, Sinclair et al. (2008) for A5959 and Biloxi, and Sadok and Sinclair (2009a) supplied data for the remaining genotypes.

Genotype	Origin	Equations of TR (mg H_2O m^{-2} s^{-1}) response to VPD	TR predicted[a] for 1.5/3.5 kPa
		Entire range of VPD	
Genotypes with linear TR to VPD response			
N01-11136[b]	NTCPR94-5157 × N96-7031	TR = 29.7 VPD − 27.4	17.2/76.7
PI471938	Plant introduction from Nepal	TR = 21.0 VPD + 1.8	33.3/75.3
N96-7031[b]	N7001 × N90-7241	TR = 20.7 VPD − 10.6	20.5/62.0

Genotype	Origin	Equations of TR (mg H_2O m^{-2} s^{-1}) response to VPD			TR predicted[a] for 1.5/3.5 kPa
		Breakpoint	VPD < breakpoint	VPD > breakpoint	
Genotypes with constant TR at high VPD					
PI416937	Plant introduction from Japan	2.19 kPa	TR = 34.1 VPD − 23.5	TR = 8.3 VPD + 33.1	27.7/62.0
Biloxi	Historical cultivar	1.90 kPa	TR = 23.0 VPD + 0	TR = 9.3 VPD + 27	34.5/59.6
A5959	Commercial cultivar	1.40 kPa	TR = 33.2 VPD + 0	TR = 5.7 VPD + 38.9	47.5/58.9
G00BP-160[b]	PI416937 × Benning	1.23 kPa	TR = 25.7 VPD − 14.0	TR = 13.6 VPD + 0.8	21.2/48.5
G00BP-59[b]	PI416937 × Benning	1.47 kPa	TR = 23.1 VPD − 11.0	TR = 8.3 VPD + 10.8	23.3/39.8
G00BP-78[b]	PI416937 × Benning	1.31 kPa	TR = 30.6 VPD − 20.8	TR = 9.1 VPD + 7.4	21.1/39.3
Benning	Hutcheson × Coker 6738	1.96 kPa	TR = 18.9 VPD − 9.6	TR = 4.7 VPD + 18.1	18.7/34.6

[a] TR ratio predicted from the equations for a VPD of 1.5 and 3.5 VPD.
[b] descendant of PI416937.

measured in the leaf chamber did not change during the short measurement period so the measures of g_{H2O} reflected the ambient values. Thus the g_{H2O} data were analyzed as a function of the ambient atmospheric VPD at the time the leaf was inserted in the leaf chamber. Ambient atmospheric VPD was calculated using the ambient air temperature, ambient relative humidity and the equation of Buck (1981).

2.3. Measurements of leaf characteristics

Leaf mass area (LMA, g m^{-2}) was measured on most leaves after each gas exchange measurement by punching a 3.89 cm^2 leaf disc from the area of the leaf that was enclosed in the gas-exchange chamber. The leaf discs were stored in coin envelopes and upon return to the laboratory dried for a day at 65 °C, and weighed. In addition, leaves were collected for measurement of leaf color using a SPAD meter (Model 502, Konica Minolta). Twenty one representative leaves of each genotype (7 replicates per plot) were collected on the last day of gas exchange measurements, placed in moistened plastic bags, stored in a cooler, and transported to the laboratory. Within 36 h the SPAD measurements were made. A preliminary experiment demonstrated that storage of soybean leaves in moistened bags in the dark at room temperature resulted in only a 7% decrease in SPAD readings after 12 days. Two SPAD readings were made per leaf and the average of these readings was used as a sample. Since the SPAD readings were intended for use as an easy-to-measure field indicator of photosynthetic capacity, calibration of the relationship between the SPAD readings and chlorophyll content was not necessary, although a strong relationship is reported to exist for soybeans (Thompson et al., 1996).

2.4. Data and statistical analysis

The genotype-specific relationship between g_{H2O} and VPD was determined as a general linear regression model (GLM) of g_{H2O} as response variable, VPD and VPD^2 as covariates, genotype as a factor and interactions were included to test for differences in slopes. These polynomial expressions allowed the relationship to be quadratic or linear as may be the case. However, the minimum adequate model (removing all non-significant interactions and predictor variables) was determined in a similar manner to that described by Gilbert et al. (2011). It should be noted that the quadratic g_{H2O} to VPD response equations used here are solely empirical, and may unduly emphasize a continuous TR to VPD response.

The relationship between A and g_{H2O} was determined for each genotype with A as the response variable and the natural logarithm of g_{H2O} as a covariate, genotype as a factor and interaction were included. As there was no significant interaction, this model could be simplified to analysis of covariance (ANCOVA) with A to $\ln(g_{H2O})$ responses of each genotype being parallel to each other, with the elevation shift as an indication of photosynthetic capacity differences as discussed in detail in Gilbert et al. (2011).

Photosynthetic capacity variation was interpreted as variation in WUE_{PC}; WUE_{PC} is the A/g_{H2O} ratio where A is the photosynthetic rate measured for a genotype at a reference g_{H2O} common to all genotypes being compared. The A value is derived from inputting the reference g_{H2O} into the regression equation of A to $\ln(g_{H2O})$ relationship for each genotype. In this way variation in photosynthetic capacity of all genotypes can be compared at a common g_{H2O}, when in fact many genotypes have limited A due to reduced g_{H2O} at high VPD. To be consistent with Gilbert et al. (2011), PI416937 was used as a reference genotype. The average percentage difference in WUE_{PC} between a genotype and the reference genotype ($\%\Delta WUE_{PC}$) was regressed against factors that may predict photosynthetic capacity, such as WUE_{PC} measured in the greenhouse, LMA, and chlorophyll content, as indicated by the SPAD readings.

3. Results

3.1. Environmental conditions

Over the period of measurement, increasing midday temperatures and in part decreasing humidity resulted in progressively higher VPD through the day, so that gas exchange measurements were taken over a large range of VPD. A 10 °C range in air temperature and a 4-fold difference VPD were observed during the 4 d of field measurements (Table 2). These conditions were similar to, or more extreme than, those previously measured in laboratory or greenhouse experiments (Table 2). Diurnal variation or short cloudy periods with moderate PPFDs did not appear to have direct effects on the measured photosynthetic rates or g_{H2O}. Specifically, the slight decreasing trend in g_{H2O} over the day was removed if g_{H2O} was detrended for the effects of changing VPD, and there was no relationship between A or g_{H2O} with ambient PPFD measured with the LI-COR external quantum sensor. That is, ambient PPFD was always sufficiently high to saturate the processes of leaf gas exchange.

3.2. g_{H2O} to VPD relationships

The slope of the g_{H2O} to VPD^2 relationship differed significantly among genotypes (genotype × VPD^2; $F_{9,163} = 3.82$, $p < 0.001$), as did the intercept (genotype; $F_{9,163} = 8.69$, $p < 0.001$; Table 3 and Fig. 1). The linear slope term and interaction with genotype were not significant, leading to VPD responses of the form $g_{H2O} = m \times VPD^2 + c$. The greenhouse observation-based classifications of genotypes into two groups corresponded to the slopes of this relationship measured in the field. Genotypes selected for a linear TR-VPD response over the entire range of VPD had the closest slopes to zero indicating that g_{H2O} responded little to VPD (Table 3). The group of genotypes defined as having a constant TR at high VPD varied in having shallow to steep slopes.

Table 2
Ranges of air temperature (T_{air}), hourly average photosynthetic photon flux density (PPFD) and vapor pressure deficit (VPD) measured during experiments in this study, or referenced studies used to investigate genetic variation in transpiration or gas-exchange responses for soybean.

Experiment	Situation	T_{air} (°C)	PPFD(μmol m^{-2} s^{-1})	VPD (kPa)
This study	Field trial	27.7–37.9	365–1668	1.1–4.4
Gilbert et al. (2011)	Greenhouse	29.5–31.6	[a]	3.0–4.0
Sadok and Sinclair (2009a)	Greenhouse	30.9 (3.07)[b]	500[c]	0.4–3.6
Sinclair et al. (2008)	Lab chamber	30 (1)[b]	1000	~0.8–3.4
Fletcher et al. (2007)	Greenhouse	34.9 (1.8)[b]	500[c]	1.1–3.7

[a] 500 W metal halide and high-pressure sodium lamps, enhancing naturally lit greenhouse PPFD by 500–600 μmol m^{-2} s^{-1} for 16 h per day.
[b] Mean (standard deviation).
[c] 75% transmission of naturally lit greenhouse PPFD (~70% of external PPFD) into plastic measurement chambers.

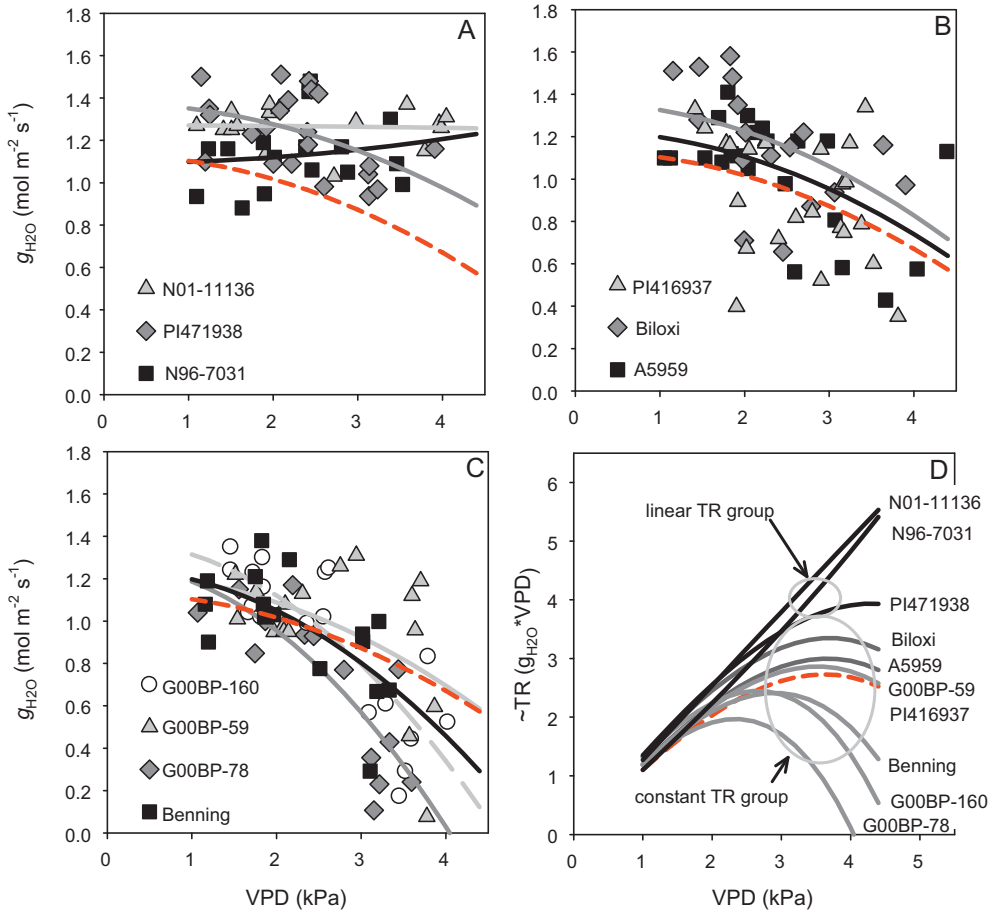


Fig. 1. Stomatal conductance to H_2O (g_{H2O}) measured under field variation in vapor pressure deficit (VPD) for ten soybean genotypes belonging to two TR to VPD response groups: linear TR to VPD responses (panel A), constant TR at high VPDs with high TR (panel B) or low TR at high VPD (panel C). The shape of the relationship between TR and VPD (panel D) was approximated by multiplying g_{H2O} by VPD for the regression shown in the other panels. Regression fits represent the quadratic model shown in Table 3 (g_{H2O} = slope × VPD^2 + intercept), and the dashed line represents the regression fit for a reference genotype, PI416937.

Inclusion of the quadratic predictor variable (VPD^2) was a significantly better predictor of the relationship than VPD (*F*-test, $p < 0.042$), indicating that there was support for non-linear g_{H2O} to VPD responses in some genotypes. Indeed, above a VPD of 3 kPa the four genotypes with the lowest TR had large variability in g_{H2O}, to the extreme of ~90% reductions in g_{H2O}. However due to this variability and fewer measurements at ~3 kPa this decreased our ability to determine the linear or non-linear nature of this response. The variation in TR with VPD implied by measured g_{H2O} relationships were estimated as TR = g_{H2O} × VPD (Fig. 1D). These relationships should be considered as illustrative, however, as the considerable leaf and canopy boundary layer effects were not taken into account. There was considerable variation between genotypes ranging from linear relationships, to genotypes with approximately constant TR at high VPD, consistent with the greenhouse data. At very high air temperature and VPD the large declines in g_{H2O} indicated a reduction in TR, a pattern not observed in greenhouse studies.

Table 3
Stomatal conductance (g_{H2O}) to vapor pressure deficit (VPD) response characteristics (g_{H2O} = slope × VPD^2 + intercept) for ten genotypes measured under natural VPD variation in a field trial. Values are means (standard errors of the mean).

Genotype	Slope (mmol m^{-2} s^{-1} kPa^{-2})	Intercept (mol m^{-2} s^{-1})
Linear TR to VPD response group		
N01-11136	−0.7 (16.2)	1.27 (0.138)*
PI471938	−24.9 (18.8)	1.38 (0.138)
N96-7031	7.2 (20.1)	1.09 (0.149)*
Constant TR at VPD group		
PI416937 (reference)	−28.9 (12.7)	1.13 (0.103)
Biloxi	−33.2 (19.2)	1.36 (0.143)
A5959	−30.5 (16.2)	1.23 (0.130)
G00BP-160	−65.0 (17.0)	1.38 (0.140)**
G00BP-59	−32.7 (17.3)	1.22 (0.148)
G00BP-78	−77.2 (20.3)	1.26 (0.164)**
Benning	−49.3 (20.4)	1.25 (0.148)

* $p < 0.10$.
** $p < 0.05$.

Using these field-measured relationships the predicted value of g_{H2O} for an arbitrary high VPD of 3.5 kPa could be compared to TR values predicted from data gathered in the greenhouse for the same VPD (summarized in Table 1). There was a strong correspondence between these values, indicating that field and greenhouse measurements represent the same process. In particular, the three genotypes with a linear TR-VPD response over the entire VPD range had the highest greenhouse TR rates and highest field g_{H2O} values (Fig. 2).

3.3. Differences in leaf photosynthetic capacity

Leaf photosynthetic rates measured in the field were generally higher than values measured in the greenhouse, and also tended to have higher variability within a genotype. In the field, large limitations on photosynthetic rates resulted from decreased g_{H2O} under varying VPD conditions (Fig. 3). For the constant TR at high VPD

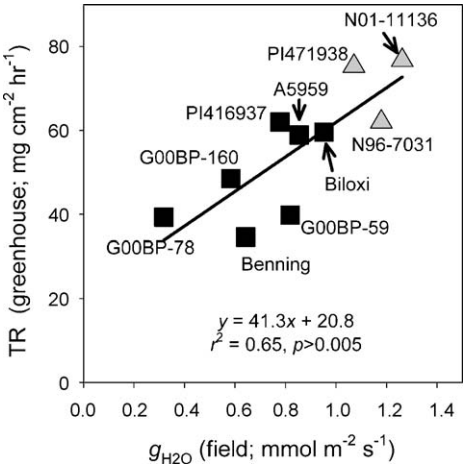


Fig. 2. Stomatal conductance to H_2O (g_{H2O}) measured in the field related to transpiration rates (TR) measured in the greenhouse. Points represent ten genotypes, where the average TR and g_{H2O} value was calculated for well watered greenhouse and field conditions of 3.5 kPa. Genotypes with linear TR responses to VPD (triangles) and those with constant TR at high VPD (squares) are shown. Greenhouse TR was estimated from previously published TR to VPD relationships summarized in Table 1, and represent high light ~1 h long measurements of plants with ~500–1000 cm^2 leaf area and were derived from references given in Table 1.

genotypes, g_{H2O} was decreased at high VPD to varying degrees, and this was reflected in larger photosynthetic limitations (Fig. 3B and C). In the case of genotypes with a linear TR-VPD response over the entire range of VPD, the lack of g_{H2O} variation with VPD resulted in little apparent decrease in photosynthesis (Fig. 3A). Although no low g_{H2O} points existed for this second group of genotypes, it was assumed that these genotypes followed A to g_{H2O} relationships of similar shape to the remaining genotypes as demonstrated previously for soil drought (Gilbert et al., 2011). When tested there was no evidence of a difference in slope for the ten genotypes (genotype × ln(g_{H2O}); $F_{9,109} = 0.57$, $p > 0.81$), but there was significant difference in A signifying a shift in the values to higher levels (genotype; $F_{9,118} = 3.58$, $p < 0.001$; Table 4). Thus genotypes had increased A relative to the A of the reference genotype PI416937 when the effects of g_{H2O} were taken into account, and therefore there was significant variation in WUE_{PC}.

The percentage genetic variability in WUE_{PC} relative to the reference genotype, PI416937, was not correlated between the field and greenhouse measurements (Fig. 4A), although the minimum $\%\Delta WUE_{PC}$ genotype (PI416937) and the maximum $\%\Delta WUE_{PC}$ genotype (N96-7031) were common between the field and greenhouse. Higher variation (~12%) in WUE_{PC} existed in the field, with

Table 4
Difference in photosynthetic rate (A) between each genotype at $g_{H2O} = 1.0$ and A (30.9 μmol CO_2 m^{-2} s^{-1}) of PI416937 genotype at $g_{H2O} = 1.0$.

Genotype	Diff. in A from PI461937 (μmol m^{-2} s^{-1})
Linear TR to VPD response group	
N01-11136	+1.9*
PI471938	+2.1**
N96-7031	+3.8***
Constant TR at VPD group	
Biloxi	+3.3***
A5959	+1.0
G00BP-160	+0.8
G00BP-59	+1.6*
G00BP-78	+0.6
Benning	+0.8

* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$ for difference with PI416937.

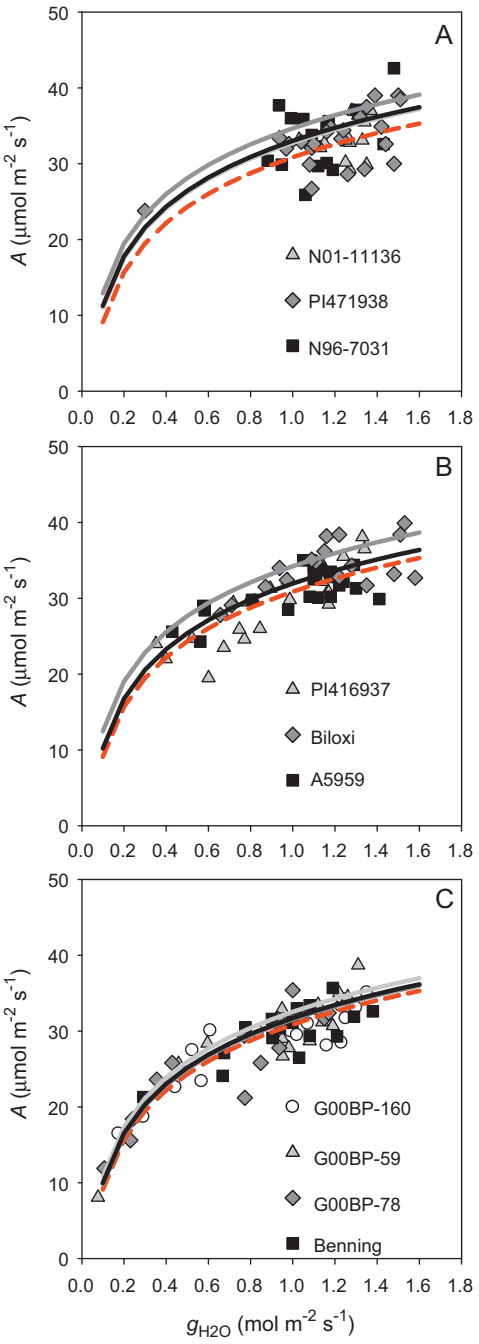


Fig. 3. Variation in stomatal conductance (g_{H2O}) induced by daily changes in VPD in the field progressively limits photosynthesis for ten soybean genotypes, from little limitation for the linear TR to VPD response group (panel A), greater limitation for the constant TR at high VPD group with high TR (panel B) or low TR at high VPD (panel C). Regression fits represent the log model shown in Table 4 ($A = m \times \ln(g_{H2O}) + \text{elevation} \times \text{genotype effect}$), and the dashed line represents the regression fit for a reference genotype, PI416937.

the tendency for genotypes with a linear TR response over all VPD to have the highest WUE_{PC}, despite variation in g_{H2O} being accounted for in this analysis (Fig. 4B). Leaf mass per unit area (LMA) was a significant predictor of WUE_{PC} (Fig. 4C), while SPAD values were not correlated (Fig. 4E). Significant genetic variability existed in LMA ($F_{9,145} = 2.39$, $p > 0.014$; Fig. 4D) and SPAD values ($F_{9,200} = 10.3$, $p < 0.001$; Fig. 4F). The investment in leaf mass per unit area was on average 8% higher in the field than the Gilbert et al. (2011) greenhouse study (paired t-test, $p < 0.007$, with nine common genotypes),

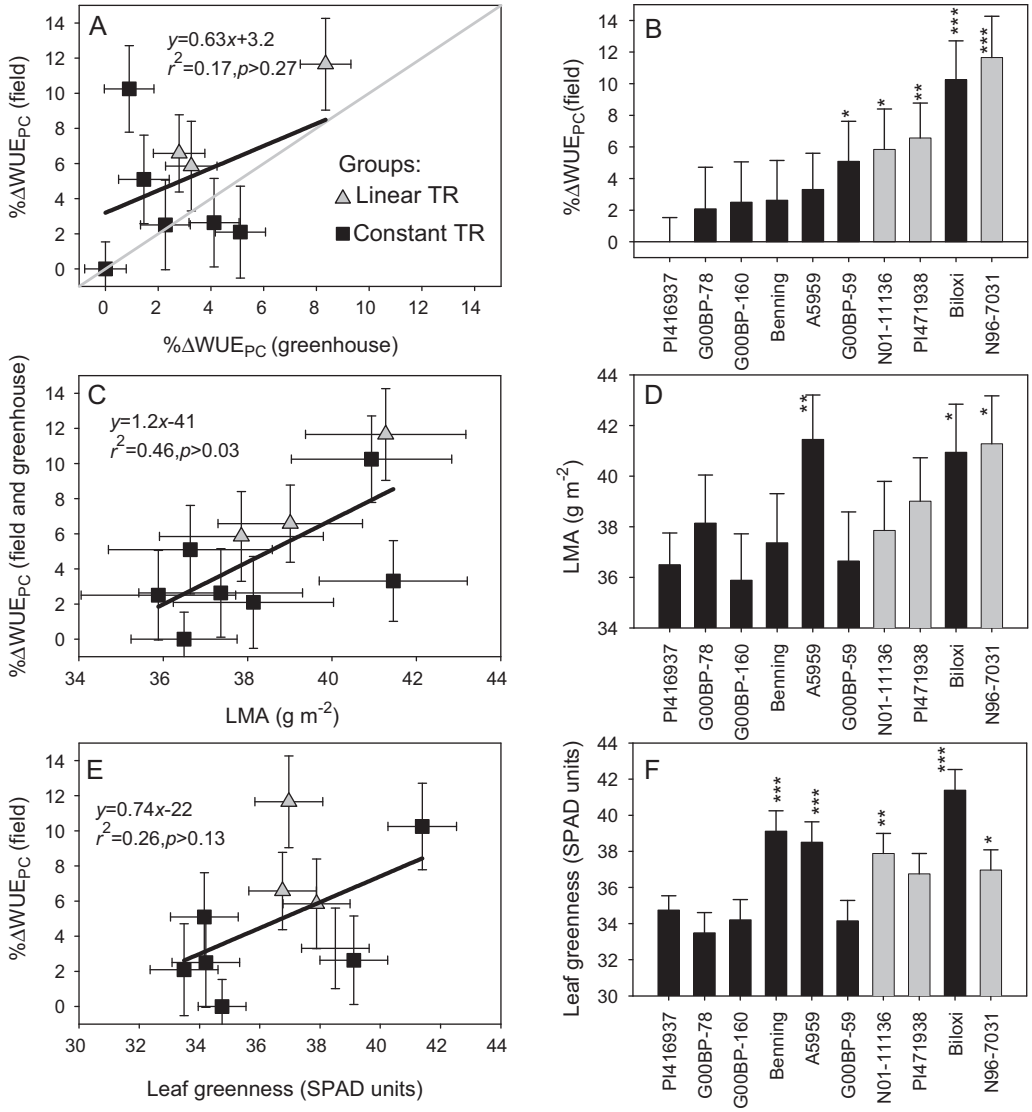


Fig. 4. Percentage genetic variability in photosynthetic capacity (%ΔWUE_{PC}) measured in the field, related to %ΔWUE_{PC} measured in the greenhouse (panel A and B), leaf mass area (LMA; panel C and D), and leaf greenness (panel E and F). *p<0.05, **p<0.01, ***p<0.001 for difference with PI416937.

and average genotype values for the field and greenhouse were correlated ($r^2 = 0.502$, $p > 0.03$).

4. Discussion

4.1. Transpiration response to VPD

Greenhouse results demonstrated large genotypic variation in whole plant TR response to increasing VPD (Sadok and Sinclair, 2009b). Most genotypes exhibited a continuously increasing TR with increasing VPD over the entire range of VPD. A few genotypes, however, had an initial linear response in TR with increasing VPD, followed by a plateau or shallow linear response to VPD above some breakpoint VPD (Sadok and Sinclair, 2009a) (Table 1). The latter genotypes may have considerable potential for increased soil water conservation and greater yields in areas prone to late season drought (Sinclair et al., 2010).

In this study, the VPD response was assessed on field-grown plants based on possible changes in stomatal conductance to water (g_{H2O}). At high VPD, the average g_{H2O} measured in the field was proportional to TR measured on whole plants in greenhouse experiments. Among genotypes, average g_{H2O} varied by 75% in this field study as compared to a variation of 55% in whole plant TR in the greenhouse study between genotypes at high VPD (Fig. 2). These results confirmed that the magnitude of transpiration in greenhouse experiments is predictive of large stomatal conductance differences in the field at high VPD. At lower VPD, g_{H2O} was more similar among genotypes, although significant differences still exist (Table 3, Fig. 1). The shapes of g_{H2O} responses to VPD varied significantly among genotypes. The responses were described empirically by the function $g_{H2O} = m \times VPD^2 + c$ with m varying from ~0 in N01-11136 and N96-7031 to a value of $-77\ mol\ m^{-2}\ s^{-1}\ kPa^{-2}$ in G00BP-78. This variation indicated that there is field variation in the TR-VPD response corresponding to the range of response shapes shown in Fig. 2D – from linear and increasing at high VPD to constant or decreasing.

There was genotypic correspondence between the field measurement of g_{H2O} response to increasing VPD and controlled environment measurements of whole plant TR response to VPD. The three genotypes chosen for increasing TR over the whole range of VPD demonstrated a similar response in g_{H2O} in the field (Fig. 1A). Those genotypes selected for little or no change in TR at VPD above a threshold exhibited in the field a decreasing g_{H2O} at high VPD reaching g_{H2O} as low as $0.1\ mol\ m^{-2}\ s^{-1}$ (Fig. 1D). These results

demonstrated an expression of greater genetic variability in the field than that measured in the greenhouse. Likely causes could be due to ambient conditions of greater temperatures (>35 °C), light (>1000 μmol m^{-2} s^{-1}) and VPD (>3.5 kPa) measured in the field (Table 2) than in the greenhouse study.

The objective of this study was to assess the correspondence in TR to VPD responses between whole plants in the greenhouse and the g_{H2O} measurements in the field. These results clearly demonstrated this correspondence for the tested genotypes. While the source of variation among genotypes in this study was due to apparent differences in leaf hydraulic limitations, a similar correspondence between greenhouse and field observations were reported by Bunce (1984) among soybean genotypes with differing root hydraulic conductance. In his study, the same ranking among genotypes in leaf-level VPD responses between the greenhouse and field was found.

4.2. Leaf photosynthetic capacity variation

Leaf photosynthetic rates varied considerably among genotypes, largely due to variation in g_{H2O} induced by VPD changes. Once this effect was de-trended through analysis of A at a constant g_{H2O} (the measure WUE_{PC}), genotypic variation in photosynthetic capacity could be analyzed. There was considerable variation in WUE_{PC} and this variation corresponded in magnitude to previously measured variation in the greenhouse (Gilbert et al., 2011). However, field genotypic differences did not correlate with greenhouse values (Fig. 4A). It seems unlikely that the reason for failure to find a correlation between the greenhouse and field results was due to differences in measurement conditions or investments in leaf mass per unit area.

Measurements of photosynthetic capacity are laborious and are necessarily limited to a very limited number of genotypes. To exploit possible genetic variation in a breeding program, it is necessary to develop indices of photosynthetic capacity that can be readily implemented for a large number of genotypes. One possibility is the dry mass per unit leaf area (LMA), or the equivalent specific leaf weight since it may relate to the resource investment in photosynthetic capacity in crops. This relationship, however, was not the case for the greenhouse experiment of Gilbert et al. (2011), but LMA was a significant predictor of genetic variability in photosynthetic capacity in the field (Fig. 4C). This is consistent with a number of studies (Wells et al., 1986; Morrison et al., 1999), while others found little consistent relationship between photosynthesis and LMA in soybean (Secor et al., 1982; Thompson et al., 1996). We note that part of this variability may be ascribed to different definitions of photosynthesis: leaf-level apparent photosynthesis and canopy apparent photosynthesis may have different relationships with LMA. LMA was not very strongly related to photosynthetic capacity differences (Fig. 4C), it is not highly heritable (Wiebold et al., 1981), and the relation between LMA and yield or photosynthesis varies over the growing season (Subrahmanyam, 2002).

The lack of variation in g_{H2O} for some of the genotypes with the highest photosynthetic capacities did not allow a direct test of whether photosynthetic capacity differences are maintained under conditions of varying g_{H2O}. However, the photosynthetic differences were maintained in previous soil drought experiments conducted in the greenhouse in which g_{H2O} varied for all genotypes (Gilbert et al., 2011). This remains to be confirmed in the field, but it seems likely photosynthetic compensation occurs.

4.3. Quantitative effects of variation in stomatal conductance and photosynthetic capacity in the field

Genetic variability in the response of stomatal conductance to VPD offers a major avenue of breeding crops for water conservation, but as demonstrated here there are consequent decreases in photosynthesis rates. What are the quantitative possibilities for breeding decreased stomatal conductance and compensating for this effect? Among the ten genotypes studied here the maximum variation in average g_{H2O} for a genotype at high VPD (3.5 kPa) was a 73% decrease, similarly at a moderate VPD of 2.5 kPa the maximum decrease was 34% relative to the average genotype at a low VPD of 1.5 kPa (Fig. 5). For these extreme decreases in g_{H2O} only 38% and 12% decreases in photosynthesis rate would occur (3.5 and 2.5 kPa, respectively), indicating that due to the curvature of the A to g_{H2O} relationship, breeding for reduced g_{H2O} at high VPD would have less effect on A than on transpiration rate (Fig. 5). However this would still result in a loss of some potential for productivity if the water conservation did not ultimately benefit A later in the growing season. The variation in photosynthetic capacity, as indicated by WUE_{PC}, was ~12% maximum variation among genotypes (Fig. 5). While moderate, this would be sufficient to compensate for the photosynthetic limitation caused by a VPD of 2.5 kPa, and about 25% of the limitation caused by a VPD of 3.5 kPa, relative to 1.5 kPa. As a bonus it would increase low VPD photosynthesis by 12%, when all genotypes had similar g_{H2O}. It should be noted that the genotypes used in this study were largely chosen for a high degree of variation in TR response to VPD, and not for differences in photosynthetic capacity. Thus, it is possible that greater genetic variation in photosynthetic capacity may exist in other genotype selections

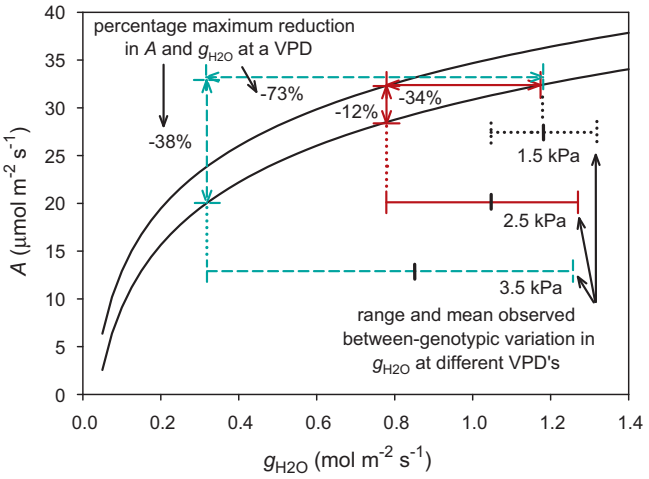


Fig. 5. The effects of genetic variability in g_{H2O} response to VPD at 1.5, 2.5 and 3.5 kPa VPD on limiting photosynthesis. The g_{H2O} range of the average values for each genotype is plotted as mean-range bars under the A to g_{H2O} relationship for PI416937 (lower curve), and the photosynthetic compensation effect of increased photosynthetic capacity as exemplified by N96-7031 (upper curve). These two genotypes were chosen as they were at the extremes of the photosynthetic capacity differences, and therefore represent the maximum practical photosynthetic compensation effect possible given the observed field variation in photosynthetic capacity. The horizontal, vertical bars and percentages represent the maximum reductions in A and g_{H2O} at various VPD.

5. Conclusions

Field measurements of leaf photosynthesis and stomatal conductance confirmed the results of whole plant measurements made in controlled environments on ten soybean genotypes. Specifically, all studied soybean genotypes had similar stomatal conductance under low VPD conditions, diverging at higher VPDs, such that some genotypes showed no reduction in g_{H2O}, and others a decrease in g_{H2O} of up to 90%. These decreases corresponded to proportionally smaller photosynthetic limitations, which could be partially ameliorated through variation in photosynthetic capacity. Varia-

tion in photosynthetic capacity and stomatal conductance has been shown to exist in greenhouse, under a range of environmental conditions, and is now demonstrated in the field. Therefore breeding water conserving soybeans through a constant transpiration rate at higher VPD represents a viable strategy for improving the potential yield of soybeans under water limited conditions.

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